

Diagnostic Capabilities of Cased Hole Logging

A Structured Framework for Identifying Common Production Problems

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Four Diagnostic Triggers Prompting Cased Hole Evaluation



Poor Initial Performance

Failure to meet expected producibility benchmarks upon completion.



Unanticipated Phase Production

Unexpected increases in water or gas volumes.



Gradual Production Decline

Slow deterioration of wellbore or reservoir conditions over time.



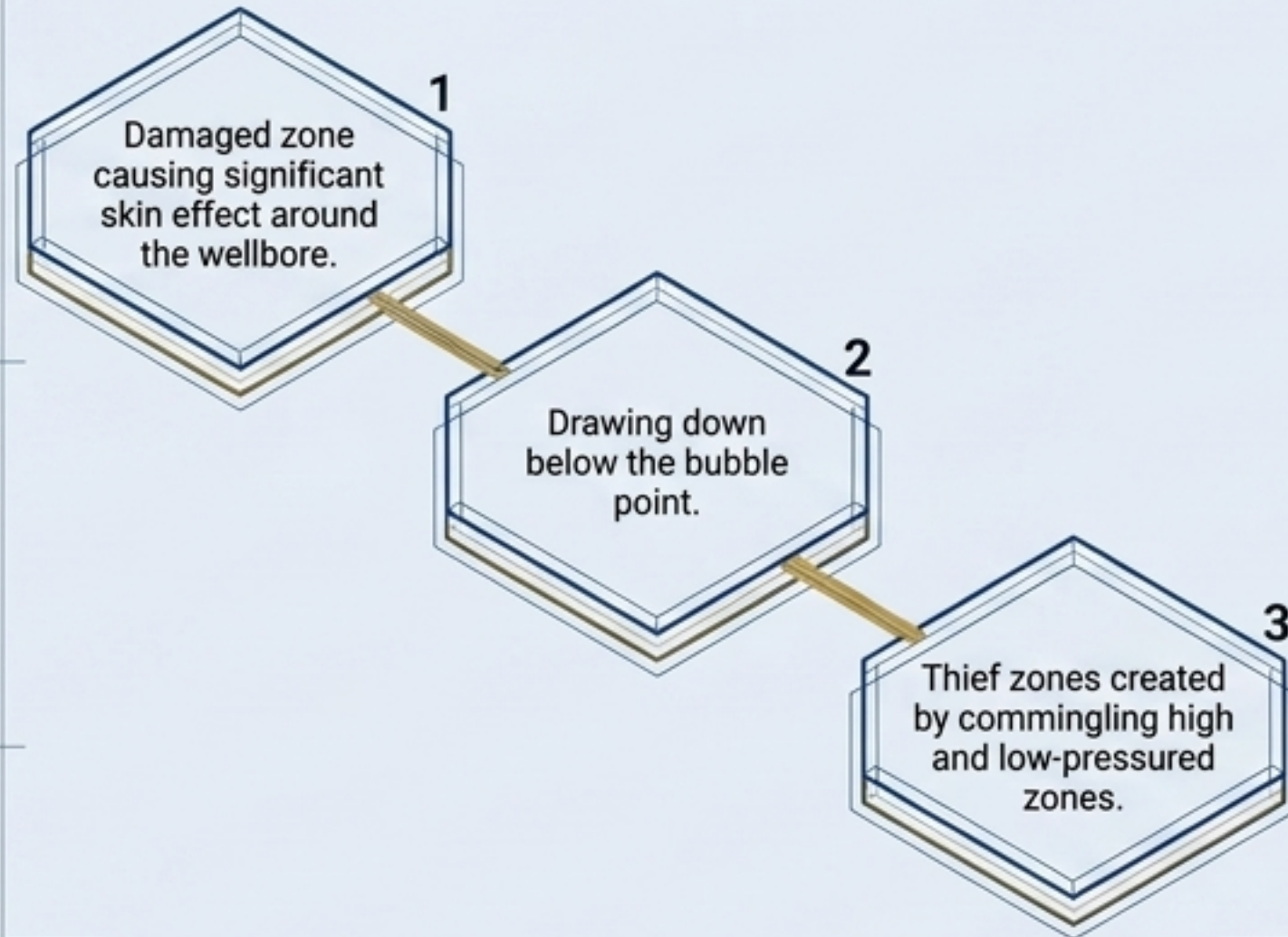
Well Integrity & Miscellaneous

Unpredictable mechanical failures or treatment evaluations.

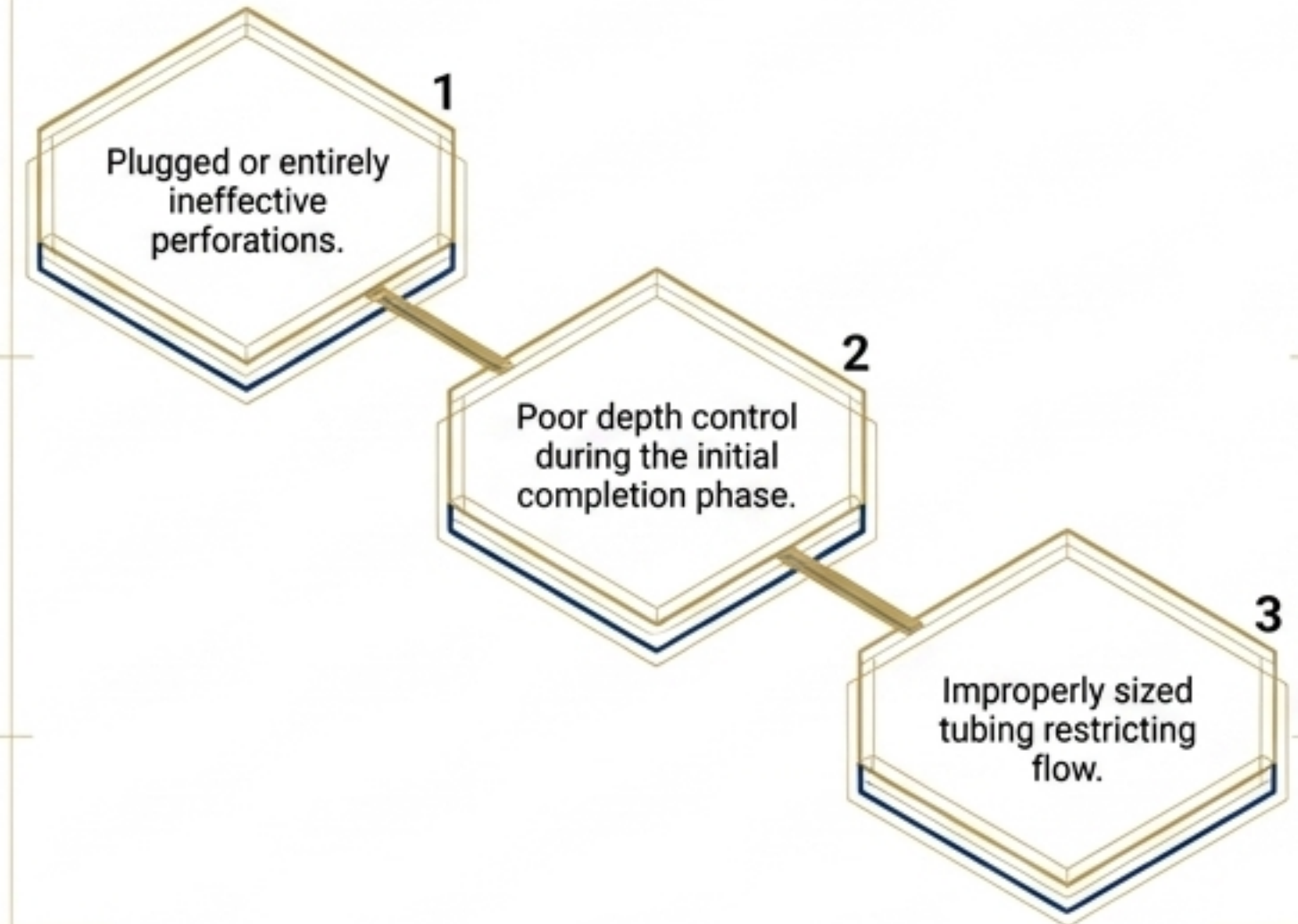
Diagnostic Trigger I: Isolating Causes of Poor Initial Performance

Poor initial performance is indicated by the well's failure to meet producibility expectations. Cased hole logs isolate whether the failure stems from the rock or the pipe.

Formation & Reservoir Dynamics



Completion & Operational Deficiencies



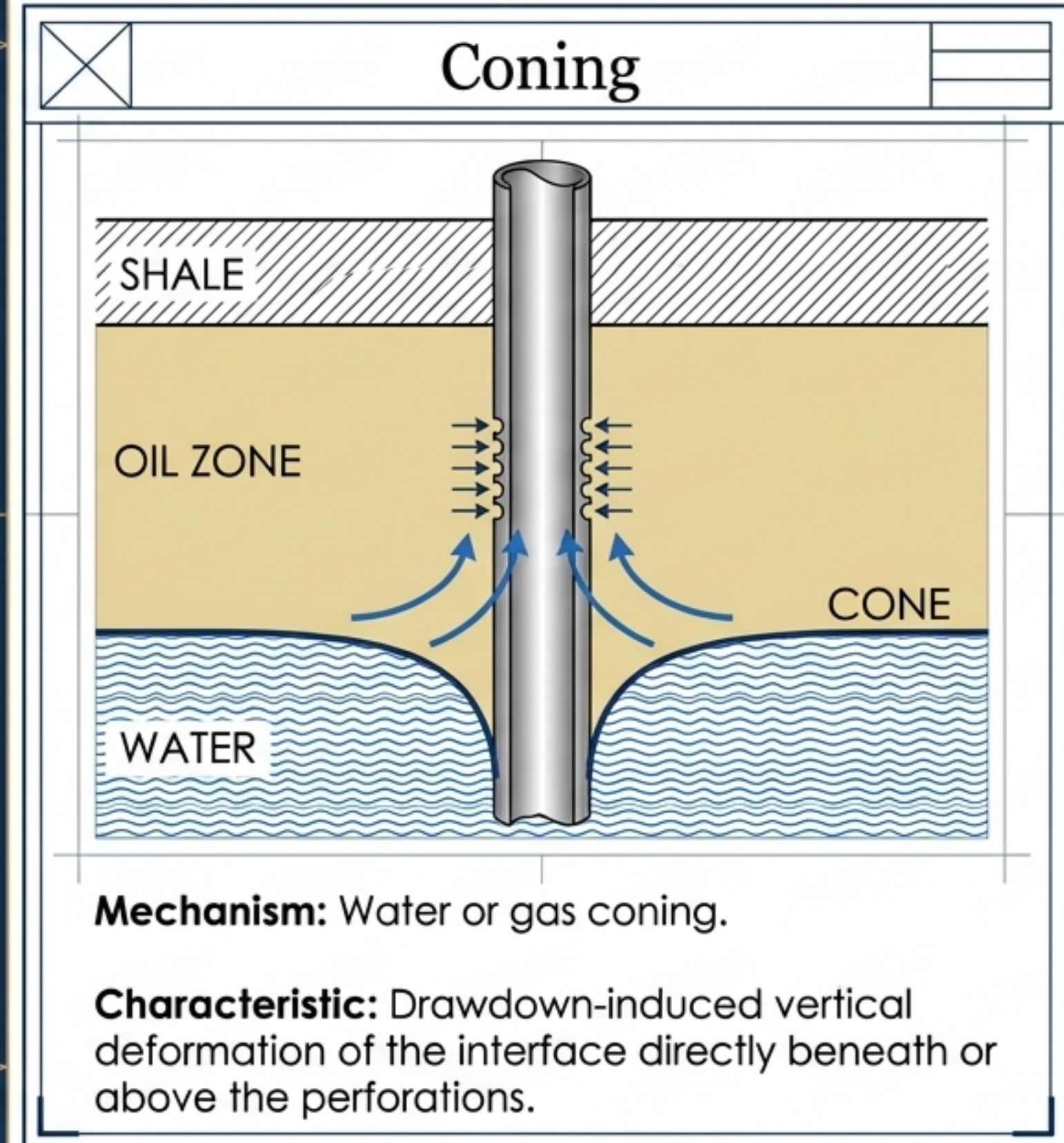
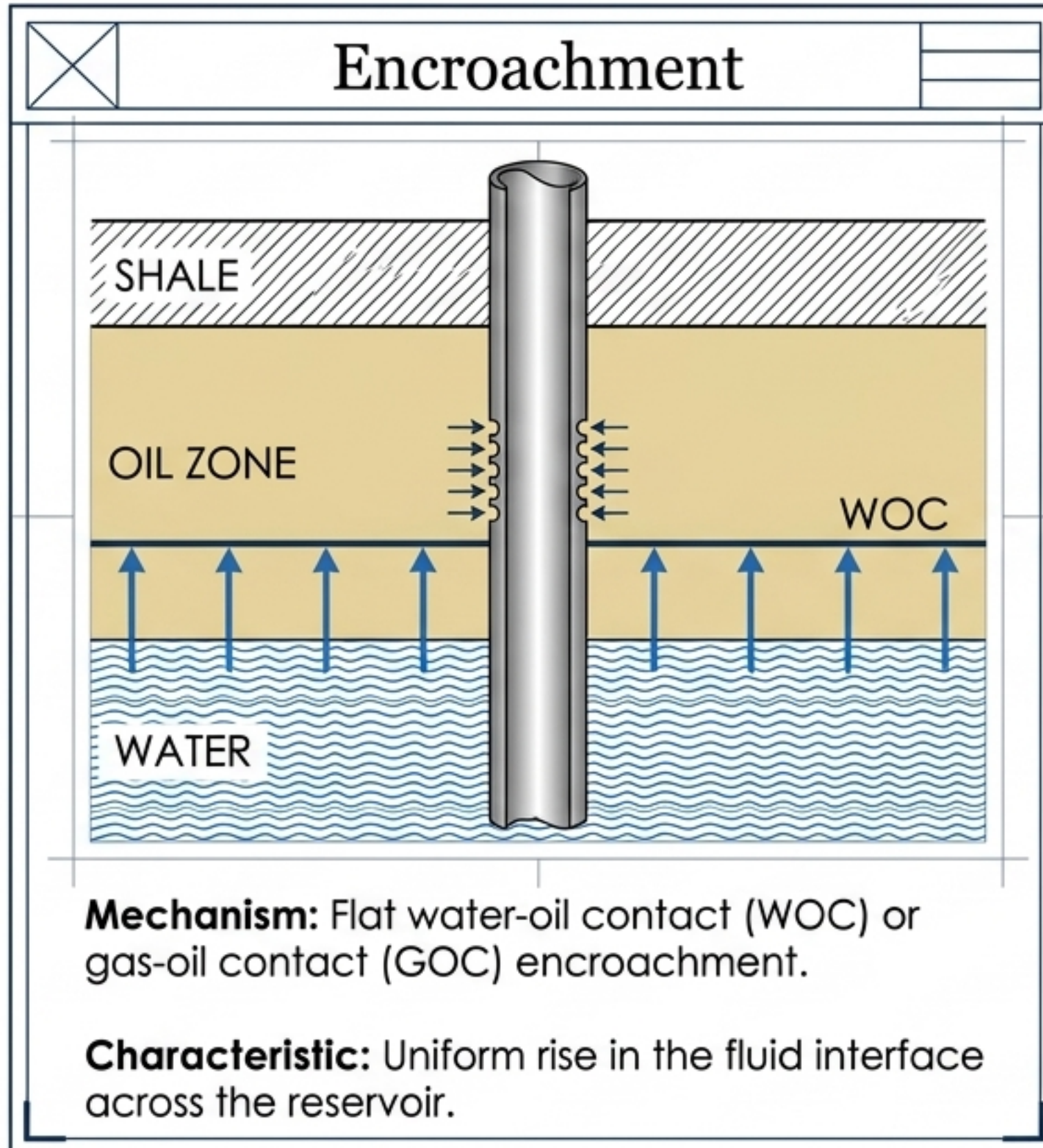
Diagnostic Trigger II: Unanticipated Phase Production

Core Issue: Unanticipated fluid production typically refers to unexpected initial production of an undesired phase, or sudden increases in water or gas production.

Monitoring of fluid contacts (WOC/GOC) via cased hole logs allows operators to anticipate changes in production before surface symptoms peak.

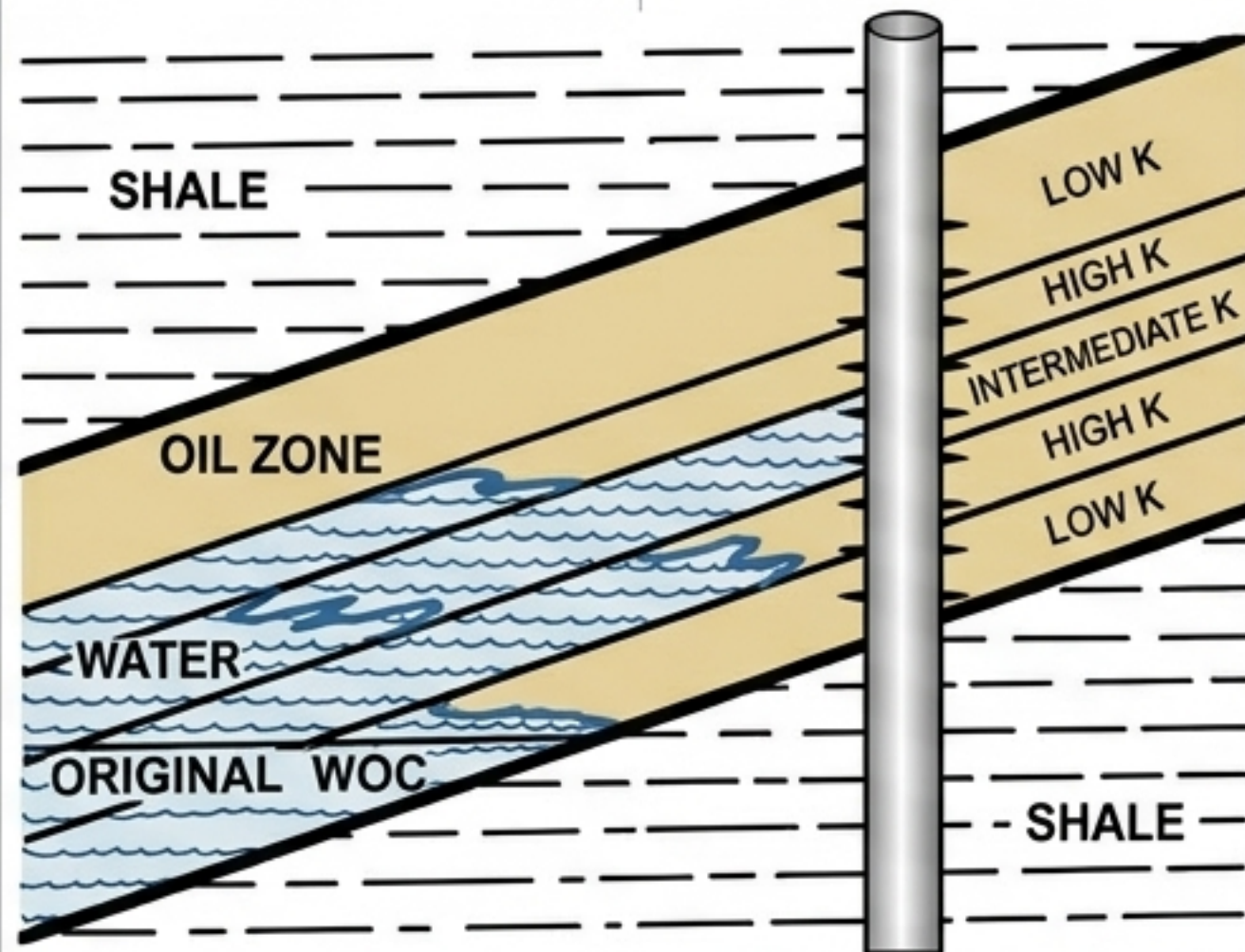


Differentiating Fluid Geometries: Encroachment vs. Coning



Differentiating Pathways: Permeability Fingering vs. Mechanical Channeling

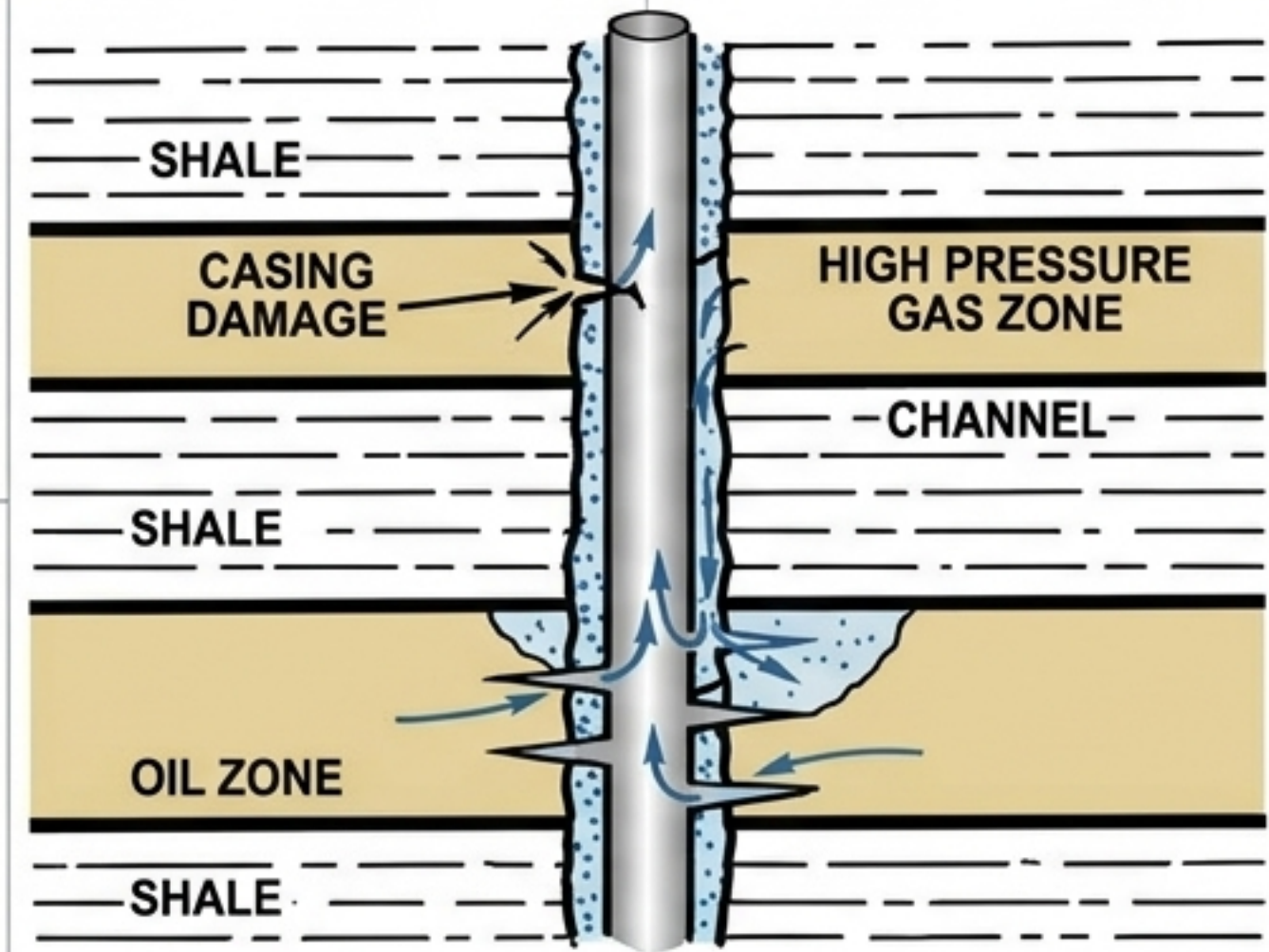
Permeability Fingering



Mechanism: Fingering through high-permeability zones.

Diagnostic Signature: Undesired fluid rapidly bypasses lower-permeability rock, breaking through specific stratigraphic layers while leaving intermediate zones unswept.

Mechanical Channeling



Mechanism: Channeling from nearby zones and casing damage.

Diagnostic Signature: Loss of zonal isolation allows fluid migration through the annulus or directly through high-pressure leaks in damaged casing.

Diagnostic Trigger III: Drivers of Gradual Production Decline

Declines in production are gradual and usually result from a slowly deteriorating condition. Time-lapse monitoring with logs sheds light on the exact driving mechanism.

Pressure & Flow Dynamics

- Reservoir pressure decreases
- Relative permeability effects altering multi-phase flow

Physical Wellbore Blockage

- Near-wellbore permeability loss due to particulate movement
- Restriction due to sand fill
- Fracture closure

Hydrostatic Changes

- Increase in hydrostatic head due to increased production of the heavy phase

Diagnostic Trigger IV: Well Integrity & Treatment Evaluation

Beyond standard production changes, cased hole logs are critical for diagnosing an uncountable number of miscellaneous structural and operational problems.

Treatment Efficacy

Evaluating the effectiveness of stimulation or remediation efforts (fracture jobs, acidizing, or squeeze cement jobs).

Solids Management

Diagnosing the location and severity of sand production entering the wellbore.

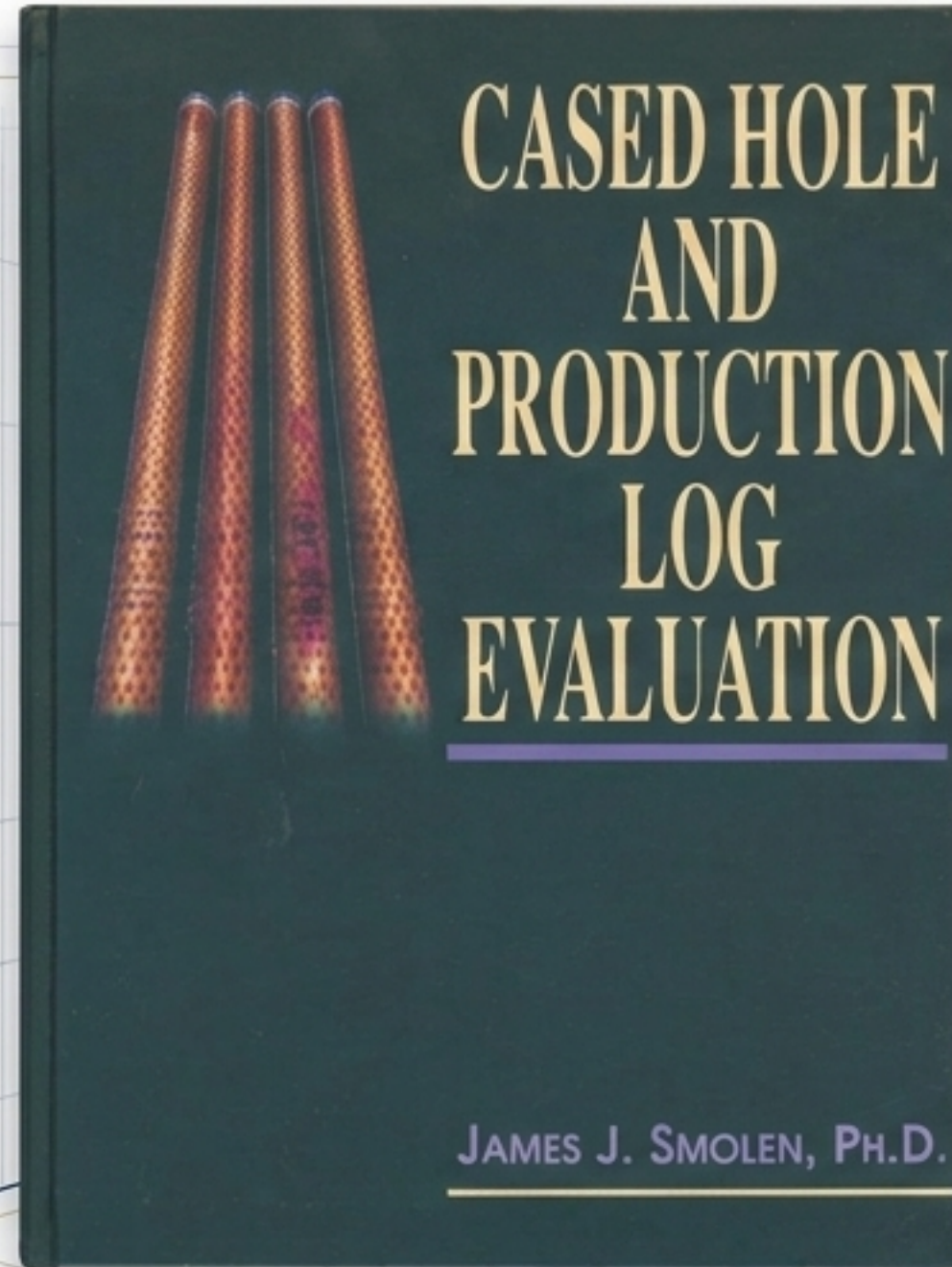
Mechanical Integrity

Pinpointing exact locations of tubing, casing, or packer leaks compromising the closed system.

The Diagnostic Blueprint: Synthesized Log Evaluation Matrix

	Reservoir / Geology	Near-Wellbore Interface	Mechanical / Tubular
Poor Initial Performance	Commingling thief zones, below bubble point.	Damaged zone / skin.	Plugged perfs, poor depth control, tubing size.
Phase Change	Encroachment, Fingering.	Coning.	Channeling, Casing Damage.
Gradual Decline	Pressure drop, relative perm.	Particulate perm loss, fracture closure, sand fill.	Increased hydrostatic head.
Integrity & Misc	N/A	Frac/Acid/Cement effectiveness, Sand production.	Tubing, casing, or packer leaks.

Source Material & Academic References



Foundational Text:

Cased Hole and Production Log Evaluation by James J. Smolen, Ph.D.

Source Authorship & Expertise:

Based on diagnostic frameworks compiled by Abbas Radhi Abbas, Chief Engineer / Petrophysics Manager (Missan Oil Company).